module-12-movement-of-sun - Darshana (60-min workshop)

IKS Curriculum - senior band

Module 12 — Movement of the Sun (Darśana)

Sanskrit: *Sūrya-gati* · **Band:** Senior (9-12) · **Duration:** 60 min · **Path:** Darśana (Workshop)

Solar year, solstices, equinoxes, and the 12 rāśis.

> **Darśana note.** This workshop condenses Module 12's full 10-day arc into one hour. For the deeper version — daily lesson plans, quizzes, assessments, slide decks — switch to **Adhyayana** (the Full Course path).

Opening hook · 5 min

Project: (a) Earth's axial tilt diagram, (b) the 12 rāśis (sidereal zodiac), (c) the modern tropical zodiac. *"Three frameworks, two ancient and one modern. They drift apart over millennia. Today: why."*

Core idea · 15 min

Earth-Sun astronomy + Indian framework.

1. **Earth's axial tilt** (~23.4°) → annual variation of Sun's declination → seasons.
2. **Uttarāyaṇa** in Indian tradition: starts at Dec solstice (modern view) OR Makara Saṅkrānti (~Jan 14, traditional view). The ~24-day gap is the *ayanāṁśa* — the precession-of-equinoxes difference between sāyana (tropical) and nirayana (sidereal) reference frames.
3. **Precession:** Earth's axis wobbles with ~26,000-year period. Discovered ~130 BCE by Hipparchus. Independently computed by Indian astronomers (Bhāskara II, ~1150 CE). Today's *ayanāṁśa* ~24°.
4. **12 rāśis** — sidereal zodiac, anchored to fixed stars. Tropical zodiac anchored to equinox. Drift apart by ~1° per 72 years.

Honest framing: the Indian framework (sidereal) and modern Western framework (tropical) are both valid astronomical conventions. They serve different purposes (Indian tradition: long-term star-anchored; modern: season-anchored).

Demonstration · 10 min

Sidereal vs tropical comparison. Use a sky-map app (Stellarium). Show where the Sun ACTUALLY is on March 21 vs where the tropical zodiac says Aries begins. Quantify the gap.

Source moment · 5 min

Direct verse. Sūrya-siddhānta 14.1:

> *bhānor varṣa-gatiḥ rāśi-dvādaśakaṁ kālaḥ sāyanam*

"The Sun's annual motion is twelve rāśis — this is the sāyana (with-ayana) year."

Note the technical precision: *sāyana* specifies the reference frame. Indian astronomy distinguished sidereal and tropical conventions explicitly.

Hands-on activity · 15 min

Precession project pitch. Each student researches one named historical observation of precession (Hipparchus 130 BCE, Bhāskara II 1150 CE, Newton 1687). Present: who? where? what did they observe?

Reflection + take-home · 10 min

"Which is more useful in modern times — sāyana or nirayana? For which purpose? Defend with evidence."

Go deeper into Adhyayana

The 10 days you're skipping if you only do this workshop:

- **Day 01** — Sun's yearly path
- **Day 02** — Axial tilt and seasons
- **Day 03** — Uttarāyaṇa and Dakṣiṇāyana
- **Day 04** — Equinoxes — viṣuva
- **Day 05** — 12 rāśis
- **Day 06** — Saṅkrānti moments
- **Day 07** — Sāyana vs nirayana — the ayanāṁśa
- **Day 08** — Precession of equinoxes
- **Day 09** — Cross-cultural calendars — solar systems
- **Day 10** — Final share + personal calendar

→ **Open the Adhyayana track for this module** to access all 10 days, the 4-audience PDFs, the slide deck, the quizzes, and the project rubric.